

4028

**B.C.A. VIth Semester (NEP)
Examination, 2026**

MACHINE LEARNING

Paper : NBCA-601

Time : 2 Hours]

[M.M. : 70

Note :- Attempt *Four* questions in all. *One* question out of two questions (10 marks each) is to be attempted from Section 'A'. Remaining *three* questions (each carrying 20 marks) are to be attempted out of eight questions given in Section 'B'.

Section-A

[10×1=10]

Note :- Attempt any *one* question.

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(1) **KG-114** Turn Over

1. Explain the different types of Machine Learning systems with suitable examples. How is Machine Learning different from traditional programming ?

Or

2. Explain the K-Nearest Neighbours (KNN) algorithm with its working steps. What are the different distance metrics used in KNN ? Discuss the effect of choosing different values of K.

Section-B

[20×3=60]

Note :- Attempt any three questions.

3. Explain the concept of Statistical Learning. What is the difference between training loss and test loss ? Discuss the bias-variance tradeoff and sampling distribution of an estimator.

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(2)

KG-114

4. Discuss Decision Trees, Entropy and Information Gain.

Construct a Decision Tree for the following dataset to predict the 'Result' :

Weather	Time	Road	Result
Sunny	Morning	Clear	Go
Rainy	Morning	Clear	Go
Sunny	Evening	Blocked	Stay
Rainy	Evening	Clear	Stay
Sunny	Morning	Blocked	Stay
Rainy	Morning	Blocked	Stay

5. Illustrate Ensemble Learning. Explain Voting Classifiers, Bagging and Pasting, Random Forest and Boosting (AdaBoost) techniques.

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(3) KG-114 Turn Over

6. State Bayes' theorem. Discuss Naive Bayes classifier. Using the following dataset, classify a new instance (Size = Medium, Color = Red) into 'Good' or 'Bad'. Show all probability calculations :

Size	Color	Class
Small	Red	Good
Medium	Green	Good
Large	Red	Bad
Small	Green	Good
Large	Green	Bad
Medium	Red	Bad
Small	Red	Good
Medium	Green	Good
Large	Red	Bad
Medium	Red	Good

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(4)

KG-114

7. Explain K-Means Clustering algorithm. Apply K-Means ($K=2$) on the following data points with initial centroids $C_1 = (2, 3)$ and $C_2 = (8, 9)$. Show cluster assignments and updated centroids after two iterations :

Point	X	Y
P1	2	4
P2	3	2
P3	1	3
P4	7	6
P5	8	9
P6	6	7
P7	9	8

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Turn Over

8. Demonstrate Curse of Dimensionality. Explain

Principal Component Analysis (PCA) with its steps.

Discuss DBSCAN clustering and how it differs from K-Means.

9. Illustrate Artificial Neural Network. Explain the

architecture of a Multi-Layer Perceptron (MLP).

Discuss different activation functions (Sigmoid, ReLU, Softmax) used in neural networks.

10. Compute the output of a neural network having

2 inputs, 2 hidden neurons, 1 output neuron for input

$X = [1, 0]$. Use sigmoid activation.

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(6)

KG-114

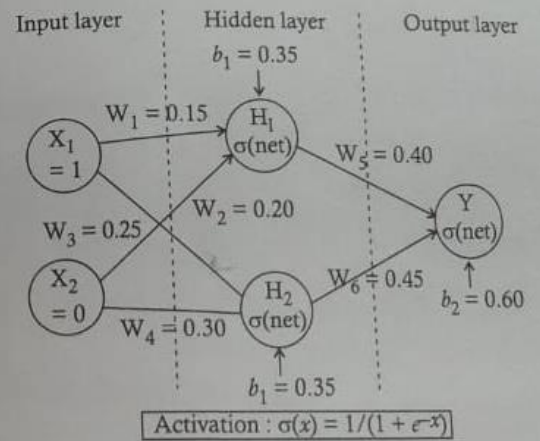
Weights :

$$W_1 = 0.15, W_2 = 0.20, W_3 = 0.25, W_4 = 0.30,$$

$$W_5 = 0.40, W_6 = 0.45.$$

Biases :

$$b_1 = 0.35, b_2 = 0.60$$



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(7) KG-114